

## Summary Report - Web QA Automation Project

For this challenge, I created a Cypress automation project using JavaScript to test some of the main interactions available on DemoQA. As challenge said, I decided not to automate everything. I focused on a few scenarios that cover different types of elements and behaviors, mainly the Practice Form and Tool Tips pages.

For the Practice Form, I covered a complete valid submission and some negative scenarios. The valid flow includes text fields, radio buttons, date picker, autocomplete, hobbies, file upload, address, State/City dropdowns, submit, and validation of the information displayed in the result modal. I also added negative scenarios for required fields, invalid email, and invalid mobile input.

For Tool Tips, I tested the hover behavior on the button, text field, and links. I also added cases to check that a tooltip is not displayed for an unrelated element and that it disappears after the mouse leaves the button.

The project uses Page Object Model to keep selectors and page interactions separated from the test scenarios. Test data is stored in fixtures, and I added a small utility to format some expected results used in the assertions, such as full name, date of birth, subjects, hobbies, picture name, and State/City.

I avoided fixed waits and preferred Cypress retry behavior and assertions based on the expected state of the application. The main test suite can be executed with `npm test`, with Chrome configured as the default browser.

I also added Mochawesome reporting and screenshots. Each execution generates an HTML report with the test results and evidence that can be used to review the run.

During the implementation, I used the official Cypress documentation as the main technical reference. When I needed to better understand specific behaviors or errors, I also researched using Google and Stack Overflow. This was especially useful for Cypress-specific behavior such as dynamic elements, hover interactions, selectors, native form validation, and reporting configuration.

One issue I found during implementation was related to native validation of the mobile field. A short mobile number behaved differently during Cypress execution compared to my manual test, so I replaced this scenario with a more stable invalid-format case using letters.

The tooltip tests also required some investigation because hover interactions are more sensitive than regular click or input actions, especially with elements that are created dynamically in the DOM.

Overall, I kept the project simple on purpose. I added reusable components only when they made sense for the current scope and avoided creating extra abstractions that were not really necessary.